

KOKAR BOGSKAR, Finland

WMO: 029790

Lat: 59.504N

Lon: 20.347E

Elev: 52

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.7	20.4	8.4	8.5	18.9	11.8	10.1	21.8	45.0	29.4	40.4	34.2	19.2	20	0.874

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.4	72.4	68.1	69.8	65.5	67.3	63.4	69.0	72.1	66.3	69.2	64.1	66.6	12.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.3	71.4	65.0	92.8	68.5	62.9	86.1	66.0	33.1	72.3	31.0	69.3	29.4	66.8	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
41.0	36.9	33.5	14.4	72.7	5.9	3.8	10.2	75.5	6.7	77.7	3.4	79.8	-1.0	82.6
			13.6	68.6	5.7	3.0	9.5	70.7	6.1	72.5	2.9	74.2	-1.3	76.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	32.3	31.5	33.1	38.5	46.8	54.6	62.2	62.7	56.5	47.9	41.4	36.9	(6)
	DBStd	11.96	5.92	5.78	5.14	3.73	4.85	4.12	4.47	3.53	3.91	4.43	4.83	5.19	(7)
	HDD50	2829	548	518	525	346	124	8	0	0	4	93	257	406	(8)
	HDD65	7176	1013	938	990	796	563	312	113	88	255	530	707	871	(9)
	CDD50	1170	0	0	0	0	26	145	379	393	199	28	0	0	(10)
	CDD65	43	0	0	0	0	0	0	27	16	0	0	0	0	(11)
	CDH74	15	0	0	0	0	0	0	12	3	0	0	0	0	(12)
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14) Wind	WSAvg	17.4	20.4	18.1	17.2	15.8	15.0	14.6	13.7	15.2	17.2	19.0	20.7	21.4	(14)
(15) Precipitation	PrecAvg	32.7	3.3	2.4	2.0	1.5	1.6	2.3	2.6	3.4	2.9	3.5	3.7	3.4	(15)
	PrecMax	43.1	7.2	4.9	3.9	3.1	3.1	4.7	5.3	7.7	5.0	5.8	6.1	6.9	(16)
	PrecMin	21.1	0.8	0.3	0.5	0.1	0.6	0.6	0.1	0.5	1.0	0.3	0.4	1.5	(17)
	PrecStd	5.8	1.5	1.1	1.0	0.8	0.7	0.9	1.3	1.7	1.1	1.3	1.5	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.8	41.6	43.8	50.7	61.9	65.6	75.5	74.3	65.5	58.9	50.9	46.8	(19)
		MCWB	41.8	38.8	39.8	43.3	54.0	58.7	68.6	70.4	62.3	56.9	49.1	45.0	(20)
	2%	DB	42.2	39.9	41.8	47.1	58.0	62.7	73.2	71.6	63.6	56.5	49.5	45.3	(21)
		MCWB	40.6	37.8	38.7	42.3	52.7	58.5	68.4	67.9	60.5	53.8	48.0	43.5	(22)
	5%	DB	41.1	39.0	40.3	44.7	55.7	61.2	71.4	68.7	62.5	54.9	48.4	44.2	(23)
		MCWB	39.8	37.5	38.0	41.3	50.9	57.6	67.4	64.3	59.4	52.7	46.8	42.4	(24)
	10%	DB	39.8	38.1	38.9	43.0	53.3	59.8	68.9	67.2	61.3	53.6	47.3	43.0	(25)
		MCWB	38.3	36.8	37.1	40.0	49.4	56.7	65.1	63.3	58.5	51.6	45.5	41.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.4	39.7	40.5	45.3	56.3	61.8	71.1	71.6	63.8	57.3	49.8	45.6	(27)
		MCDB	43.1	40.6	42.4	49.2	59.7	64.1	73.0	73.8	65.3	58.6	50.2	46.5	(28)
	2%	WB	41.3	38.6	39.4	43.2	53.9	60.0	69.6	68.5	61.2	54.6	48.4	44.1	(29)
		MCDB	42.1	39.4	41.0	46.0	57.1	61.9	72.6	71.0	62.9	55.7	49.2	45.0	(30)
	5%	WB	40.0	37.8	38.4	41.7	51.7	58.6	67.7	65.7	60.1	53.4	47.1	42.9	(31)
		MCDB	41.1	38.7	39.9	44.2	54.9	60.5	71.4	68.0	61.9	54.7	48.3	44.1	(32)
	10%	WB	38.6	37.0	37.3	40.4	50.1	57.3	65.4	64.1	59.1	52.1	45.8	41.6	(33)
		MCDB	39.8	38.1	38.7	42.5	52.9	59.4	68.7	66.3	61.0	53.4	47.2	42.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	4.4	4.3	4.6	4.9	4.5	3.7	3.4	3.6	3.9	4.0	4.4	(35)
		MCDBR	4.7	4.3	5.5	7.3	8.6	6.2	5.0	3.8	3.8	4.2	3.8	4.6	(36)
	5% WB	MCWBR	5.0	3.9	3.9	4.3	5.3	4.6	5.2	4.2	4.1	4.3	4.3	5.0	(37)
		MCWBR	4.7	4.4	5.3	6.9	7.7	5.5	4.5	3.5	3.5	4.2	3.6	4.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.287	0.320	0.337	0.373	0.368	0.361	0.390	0.381	0.350	0.337	0.313	0.271	(40)
		taud	2.217	2.297	2.347	2.343	2.433	2.492	2.434	2.442	2.509	2.410	2.233	2.128	(41)
	Ebn at Noon		160	218	253	263	273	277	265	258	249	211	145	116	(42)
		Edh at Noon	14	22	28	33	32	31	32	30	24	19	14	10	(43)
(44) All-Sky Solar Radiation	RadAvg		81	252	750	1319	1811	1986	1860	1457	911	385	110	54	(44)
	RadStd		10	43	82	138	162	118	149	119	103	56	25	7	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.61	N/A	N/A	N/A	N/A	N/A	-189	N/A	N/A	N/A

Nomenclature: See separate page